

28 Transition elements

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Transition elements have atoms or stable ions with incomplete d subshells. Variable oxidation states, coloured complexes and catalysis arise from d-electron behaviour. Ligands donate electron pairs to a central ion; ligand substitution can change colour and coordination geometry. Colour depends on splitting and absorption of visible light; d^0/d^{10} ions are often colourless.

Worked example: follow the reasoning

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ exchanges ligands with excess NH_3 to form a deep blue ammine complex; coordinate bonds form from donor lone pairs.

Quick check · multiple choice

A Which ion has incomplete d subshell?

[1]

- A Zn^{2+}
- B Sc^{3+}
- C Cu^+
- D Cu^{2+}

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why Zn^{2+} is usually colourless and $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ coloured.

[3]

Answers and explanation

Quick check: D. Cu^{2+}

Question 1 [3 marks]

Zn^{2+} has full $3d^{10}$ so no d-d transitions [1]; Cu^{2+} has incomplete 3d subshell [1]; ligand-field splitting allows visible photon absorption [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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